Statement of Purpose for Stanford, MS in Symbolic Systems
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The mind edges between order and chaos. It is creative, adaptable, and plenty fallible; cached with memories, structured by beliefs, and reinforced through experience. These dynamics of the human brain have largely driven my interests in the artificial mind, and I became eager to explore the computational underpinnings of learning, inference, and reasoning. Ultimately, this led me to pursue an individualized degree in **mathematics and neuroscience**, allowing me to build strong foundations in probability theory, dynamical systems, statistical learning, machine learning (ML), cellular neurobiology, cognition, perception, and bio-inspired computing. There is absolutely no program that fits my interdisciplinary interests better than Symbolic Systems at Stanford, and working alongside top researchers and leaders will drive forth the success of my future endeavors within computational cognition.

Aim: I am highly interested in developing cognitive-inspired models to improve interpretability and explainability in ML. Predominantly, ML systems operate as black-box models which, unlike humans, are incapable of performing inductive leaps from data and discerning causal relationships at the concepts level. These challenges have motivated my research interests in:

1) **Causal Reasoning:** What are the symbolic underpinnings of causal reasoning (perhaps even analogical reasoning), and how can we integrate them into ML architectures to improve our understanding of the underlying causal structures in data?
2) **Model Agnostic Interpretability:** How can we disentangle convoluted ML modules into collections of more interpretable units to improve transparency (potentially leveraging insights from neuroscience via hierarchical modeling, modularity, sparse coding, etc.)?
3) **Bayesian Models**: How can we leverage Bayesian frameworks to improve generalization, adaptability, and efficiency?

Research Experience:
Machine Learning: While humans have the ability to learn concepts from few examples, ML algorithms require complex architectures that struggle to learn from minimal data. Hence, a significant challenge is to understand how a system with minimal complexity can acquire new concepts from only a single training example per class. In the LaRC lab under Dr. Kenneth J. Kurtz, my honors thesis focused on an **unsupervised**, cluster-based framework for **one-shot learning** on the Omniglot dataset to encode higher level representations of visual concepts using **Gaussian Mixture Models (GMMs)**. Addressing both classification and generative tasks, my work proposed the **Abstracted Gaussian Prototype (AGP),** which models each concept as an ensemble of inferred Gaussian components corresponding to distinct topological subparts. Inspired by Tversky's 1977 work on human category judgements of prototypes, I developed a **cognitive-inspired similarity metric** for **one-shot classification**, achieving comparable performance to leading neural approaches (~95.1% on 5-way tasks). For one-shot generation, I formulated an AGP-VAE pipeline using **variational autoencoders (VAEs)** to learn a continuous latent space to generate new creative prototype variants. My model's outputs were evaluated by human judges through a "visual Turing test" and were close to indistinguishable from human drawings (~52.25%; ideal is 50%). I successfully **defended my thesis** before a three-person committee with **highest honors.** My work is currently under review as a **full paper** at **ICML '24** as **first author,** and submitted to the **CogSci '24 Conference** as a **poster**.

Cognitive Neuroscience: At the **California Institute of Technology**, I worked under professor Shinsuke Shimojo as a SURF fellow to investigate **multi-modal perception in visual processing**. Our brains are remarkable inference machines: Even with the lack of physical sensory input, such as scotomas in the retina devoid of photoreceptors, we have the ability to interpolate missing information across the blind

region to maintain a coherent visual experience. My research investigated how ***audition* can evoke visual experiences** within these supposedly blind scotoma regions. I developed a series of audio-visual experiments in Python and worked with a **high precision eye-tracker** to map participants' scotomas and monitor pupillometry responses. The **results were striking**: **auditory stimulus can generate illusory flash perceptions within scotomas**. Notably, low-vision individuals displayed greater susceptibility to these illusions, potentially attributed to their heightened reliance on other modalities to compensate for diminished visual acuity. My work was selected for an **oral presentation** at the **2023 Caltech summer research seminar.**

Robotics: In the AIR Lab under Dr. Shiqi Zhang, I developed a novel multi-modal **human-robot collaboration** (HRC) system integrating **augmented reality (AR), eye gaze, and natural language**. Confronted with uncertainties in human communication modalities, such as saccades in gaze and errors in speech, my work proposed a decision-theoretic framework to dynamically **model hidden states of human intent** for **decision making under uncertainty** tasks using a **partially observable Markov decision process (POMDP)**. My research focused on a visual-dialogue grounding approach, resulting in a joint decision policy that robustly updates belief states for optimal sequential decision making. I demonstrated my system with human participants collaborating with a UR5e arm. This work resulted in a **first author IEEE ICRA '23 CoPerception Workshop paper**. I extended this work to my **first publication**, recently accepted at the **ACM/IEEE HRI '24 Conference**.

Creativity: My interests in computational creativity led me to first explore **human creativity**. In the LCCP lab under Dr. Steven Lynn, my research assessed creativity in 324 participants through a creative personality scale (CPS), and a verbal association task (DAT), which **investigates divergent thinking based on psychological semantic distances between word embeddings**. Investigating their correlations with cognitive, personality, and affect variables, the results surprisingly revealed weak correlations between the two measures, suggesting that divergent creativity may not neatly align with the broader construct of creative personality. These findings further sparked my interests in the multifaceted nature of creativity, and I had the opportunity to present at the **2023 APS poster conference** as **first author**.

My interests are highly aligned with professors **Dan Yamins, Noah Goodman, Jiajun Wu,** and **Tobias Gerstenberg**. My overarching goal is to develop new ways to represent various cognitive landmarks (i.e., causal reasoning, analogical reasoning, divergent thinking, etc.) and brain processes from an information theoretic perspective to inspire more interpretable, efficient, and generalizable ML architectures. (Please see attached file in "additional information" section for more detailed potential research directions in each lab). In alignment, Symbolic Systems cultivates an exceptional community invested in the significance of minds and machines. I hope to be a part of it. Thank you.